

LAPLACE TRANSFORM

Time (t) $\longrightarrow$ Frequency (s)

❖ Definition:

$$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

$$e^{-\infty} = 0$$

❖ Basic formula:

$$(i) \quad \int e^{Ax} \sin Bx \, dx = \frac{e^{Ax}}{A^2 + B^2} [A \sin Bx - B \cos Bx]$$

$$(ii) \quad \int e^{Ax} \cos Bx \, dx = \frac{e^{Ax}}{A^2 + B^2} [A \cos Bx + B \sin Bx]$$

❖ Laplace Transform of standard function:

$$(i) \quad L\{1\} = \frac{1}{s}$$

$$(ii) \quad L\{e^{at}\} = \frac{1}{s - a}$$

$$(iii) \quad L\{e^{-at}\} = \frac{1}{s + a}$$

$$(iv) \quad L\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$(v) \quad L\{\sinh at\} = \frac{a}{s^2 - a^2}$$

$$(vi) \quad L\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$(vii) \quad L\{\cosh at\} = \frac{s}{s^2 - a^2}$$

$$(vi) \quad L\{t^n\} = \frac{n!}{s^{n+1}} \text{ or } \frac{n!}{s^{n+1}}$$

$$(ix) \quad L\{e^{at}\} = \frac{1}{s - a \log C}$$

❖ Trigonometric Formulas:

$$(i) \quad \sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$(ii) \quad \cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$(iii) \quad \cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

$$(iv) \quad \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\begin{aligned}
 \text{(v)} \quad & \sin(-x) = -\sin x, & \cos(-x) &= \cos x \\
 \text{(vi)} \quad & \sin^2 x = \frac{1 - \cos 2x}{2}, & \cos^2 x &= \frac{1 + \cos 2x}{2} \\
 \text{(vii)} \quad & \sin^3 x = \frac{1}{4}[3 \sin x - \sin 3x], & \cos^3 x &= \frac{1}{4}[3 \cos x + \cos 3x] \\
 \text{(viii)} \quad & \sinh x = \frac{e^x - e^{-x}}{2}, & \cosh x &= \frac{e^x + e^{-x}}{2}
 \end{aligned}$$

EXTRA:

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$1 + \sin t = \left[\cos\left(\frac{t}{2}\right) + \sin\left(\frac{t}{2}\right) \right]^2$$

❖ Types:**(1) First Shifting Property: [e]**

If,

 $L\{f(t)\} = f(s)$, then , $L\{e^{at} f(t)\} = f(s - a)$ $L\{e^{-at} f(t)\} = f(s + a)$ **(2) Particular Value: $\left[\int_0^\infty \right]$**

$$\int_0^\infty e^{-st} f(t) dt \quad [\text{compare given question with this.}]$$

(3) Multiplication by 't' or Multiple power of 't': [t f(t)]

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} [f(s)]$$

SHORTCUTS

$$\frac{d}{ds} \left(\frac{u}{v} \right) = \frac{v \frac{du}{ds} - u \frac{dv}{ds}}{v^2}$$

$$\frac{d}{ds} \left(\frac{1}{f(s)} \right) = \frac{-1}{f(s)^2} \cdot \frac{d}{ds} [f(s)]$$

Deno. → Same → Minus → Squ.

Deno. → Dvt. → Minus → Squ.

(4) Division by 't' or Divisible by 't': $\left[\frac{f(t)}{t} \right]$

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty f(s) ds$$

(i) $\int \frac{f'(x)}{f(x)} dx = \log f(x)$

(ii) $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$

TIMEPASS:

1) $\log m^n = n \log m$ or $n \log m = \log m^n$

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2) $\log a - \log b = \log \frac{a}{b}$

3) $\left[\log \frac{a}{b} \right]^\infty = 0$

4) $-\log \frac{a}{b} = +\log \frac{b}{a}$

5) $\tan^{-1} \infty = \frac{\pi}{2}$

6) $\frac{\pi}{2} - \tan^{-1} \frac{a}{b} = \tan^{-1} \left(\frac{b}{a} \right)$ or $\cot^{-1} \left(\frac{a}{b} \right)$

7) $\tan^{-1} A + \tan^{-1} B = \tan^{-1} \left[\frac{A+B}{1-A.B} \right]$

$$\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left[\frac{A-B}{1+A.B} \right]$$

(5) **L. T. of integral:**

$$L \left\{ \int_0^t f(u) du \right\} = \frac{1}{s} \times f(s)$$

(6) **L. T. of derivative:**

$$L \left\{ \frac{d}{dt} [f(t)] \right\} = s \cdot f(s) - f(0)$$

$$L \{ f''(t) \} = s^2 f(s) - s^1 f(0) - s^0 f'(0)$$

(7) Change of scale property:

If,

$$L\{f(t)\} = f(s),$$

$$L\{f(at)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$$

(8) Second shifting property:

If,

$$g(t) = 0 \quad 0 < t < a$$

$$= f(t - a)$$

$$\text{then } L\{g(t)\} = L\{f(t - a)\}$$

$$= e^{-as} f(s)$$